H2O software

Programming languages: The H2O software has a programming interface language like Scala, R (3.0.0 or later), Java (6 or later) and Python (2.7.x, 3.5.x), etc.

Operating systems: The H2O software conventionally runs on operating systems like Mac OS X (10.9 or later), Linux (Ubuntu 12.04, RHEL/CentOS 6 or later) and Microsoft Windows (7 or later).

Graphical user interface (GUI) and browsers: Its GUI is in sync with browsers like Internet Explorer (IE10), Firefox, Chrome and Safari.

Architecture

The H2O software stack: Figure 1 represents the dissimilar elements that work together to form the H2O software stack. Within the network cloud, the figure is divided into two segments -- the top and bottom segments. The dissimilar REST API clients that exist for H2O are shown in the top segment. The dissimilar elements that run within a H2O JVM process are represented in the bottom segment. Each surface in the other segment of the diagram represents the constant colour but shows the user interface customer algorithm code as grey.

REST API clients: Through a socket connection, all REST API clients communicate with H2O.

(i) *JavaScript*: JavaScript uses the benchmark REST API, and the implanted H2O Web user interface is composed in JavaScript.

(ii) *R:* Users can compose their own R operations that run on H2O with 'ddply' or 'apply'. An R script uses the library package H2O R ['library (h2o)'].

(iii)*Python:* An H2O client API for Python has been designed. Python scripts directly use the REST API.

(iv)*Excel:* The H2O Microsoft Excel worksheet permits the user to "…import large data sets into H2O and run calculations like GLM (Generalised Linear Model) straight from Excel."

(v) *Tableau:* By using conceptualisation in Tableau, users can pull outcomes from H2O.

(vi)*Flow:* Flow is the scratch pad style Web UI for H2O.

***(a) JVM components*:** H2O software comprises a large number of nodes and each one consists of a single JVM operation. The JVM operation is divided into three branches -- core infrastructure, language and algorithms.

The evaluation expression engine for R and the Shalala Scala layer dwell under the language layer. For the customer front-end, the R evaluation layer goes about as a captive to the R Rest. Anyway, the Scala layer is the superior resident using which we can compose unique projects and algorithms of H2O.

At this layer, algorithms for H2O are automatically provided. Algorithms for significant data sets, math and machine learning algorithms for prediction and the scoring engine as well as GLM can be parsed. At this level, the core layer controls memory, the CPU and resource management.

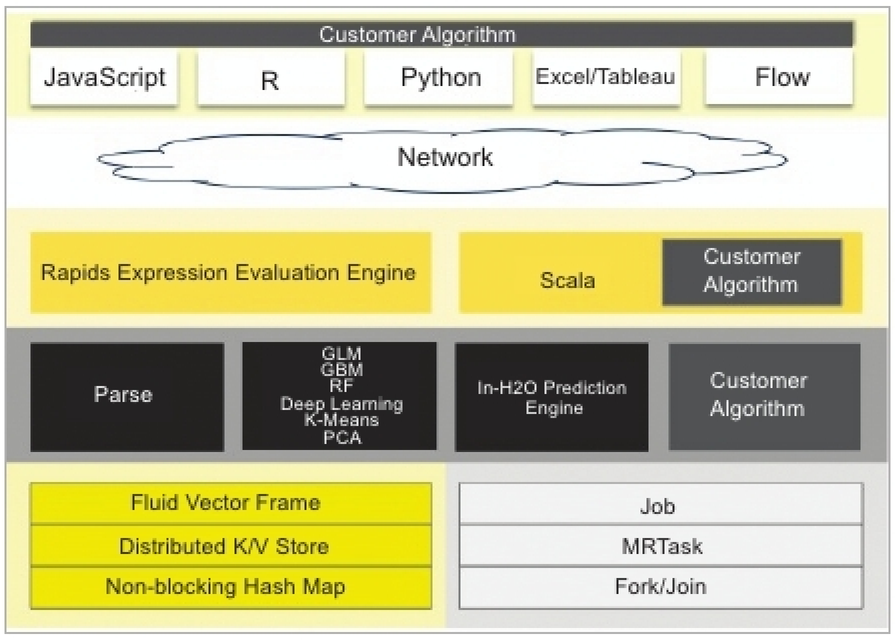

Figure 1: H2O architecture

(b) How H2O interacts with Python and R: The R script is a REST API user for the H2O cluster. H2O allows the R client to charge an H2O cluster from the R script in which the data does not all flow through R.

(i) How H2O ingests data to R scripts: The following three stages show how an R plan tells an H2O cluster to study information from HDFS into a distributed H2O frame.

Step 1: The R client calls the *importFile* function.

Step 2: The R client advises the group to read the information. In Figure 3, the arrows show the control information.

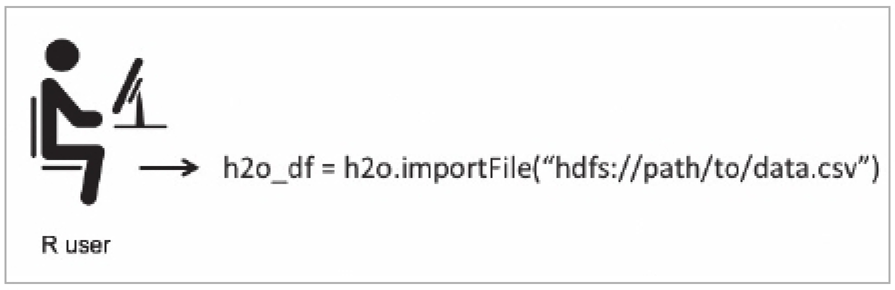

Figure 2: R user calls the *importFile* function

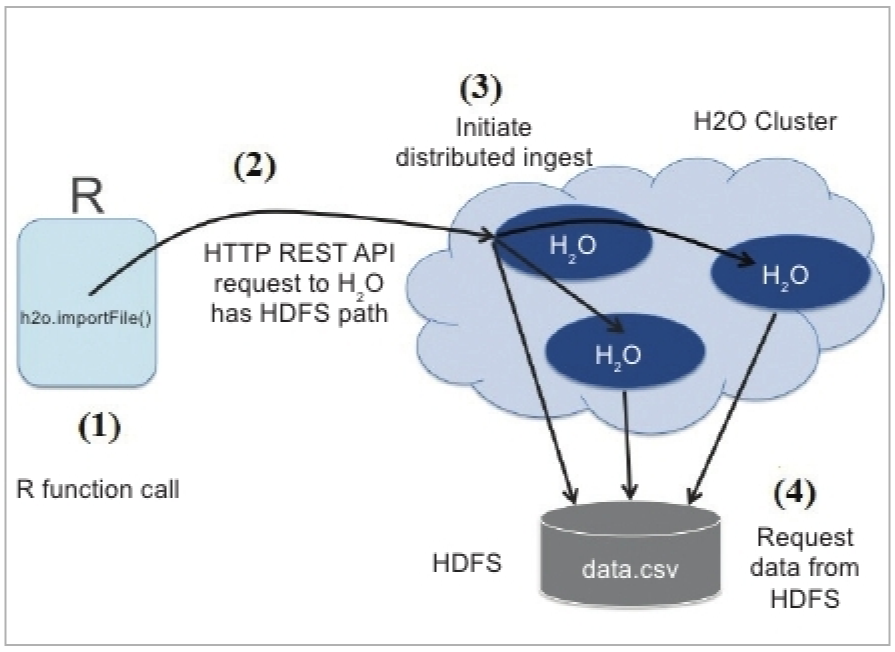

Figure 3: R client advises the group to read the information